UNIT 3. ENERGY TRANSFORMATION

3.1. CELLULAR METABOLISM

By the end of this section, the student will be able to:

- **Define cellular metabolism.**
- **Explain anaerobic and catabolic pathways in cellular metabolism.**

3.2. PHOTOSYNTHESIS

By the end of this section, you will be able to:

- **Define photosynthesis.**
- **Explain the process of dark and light reaction of photosynthesis.**
- **Describe the external and internal structure of a leaf.**
- **Describe the different chlorophyll pigments involved in the absorption of light energy.**
- **Analyze an absorption spectra of the chlorophyll pigments from the graph.**
- **Explain the roles of grana and stroma in the process of photosynthesis.**
- **Describe cyclic and non-cyclic photophosphorylation.**
- **Show an absorption spectra of chlorophyll a and chlorophyll b using graph.**
- **Explain the process of Calvin cycle.**
- **Compare cyclic and non-cyclic photophosphorylation.**
- **Differentiate between C3, C4 plants, and CAM Plants.**
- **Justify the reason why the rate of photorespiration is less in C4 plants as compared to C3 plants.**
- **Summarize the process of photosynthesis using chemical equation.**
- **Conduct an experiment to show the release of energy during photosynthesis.**
- **Conduct an experiment to show the presence of starch in green leaves.**
- **Explain the significances of photosynthesis.**

3.3. Cellular Respiration

At the end of this lesson, the student will be able to:

- **define cellular respiration**
- **describe the process of glycolysis.**
- **list the products formed at the end of glycolysis.**
- **describe the structure of mitochondria.**
- **explain how energy is harvested in aerobic respiration.**
- **describe the different stages of aerobic respiration.**
- **compare and contrast aerobic and anaerobic respiration.**
- **write the chemical equation of aerobic respiration.**
- **state the products of alcoholic fermentation by yeast.**
- **show the mechanism electron transport system in mitochondria.**
- **explain the difference between substrate-level phosphorylation and**